

EDUCATION

University of Notre Dame

PhD, Mathematics

May 2013

MS, Mathematics

January 2011

Indiana University South Bend

BS, Applied Mathematics

May 2008

EXPERIENCE

DOI, Office of Natural Resources Revenue, Denver, CO

Statistician (Data Scientist), Analytics and Compliance Planning

Spring 2022 - Present

GS-1530-13-04

- Served as ACP's statistical expert, creating products and providing authoritative guidance across ONRR on the application of statistical methods, modeling, simulation, principles, and techniques, particularly as applied to **detecting oil and gas royalty compliance risk**.
 - Instituted a cohesive road map for a suite of statistical models to be used in the detection, prediction, and deterrence of payor noncompliance and its associated monetary risks.
 - Educated colleagues across ONRR on modern statistical techniques (linear regression, logistic regression, k-means clustering, decision trees, random forests, XGBoost, neural networks, sampling methodology, population statistic estimation, general machine learning concepts), ranging from intuitive conceptual discussions to technical presentations, depending on the audience's background and needs.
 - Analyzed modern audit and compliance studies in mainstream research journals and peer agency publications and communicated summaries and potential applications to ACP and R&A leadership.
 - Translated complex findings and limitations into concise communications to empower leadership and colleagues in their effective decision-making.
- Provided industry/academic perspective on the most effective use of modern analytics tools.
 - Frequently liaised with IT to ensure successful alignment of R&A analytics goals with IT strategy and software implementation plans.
 - Advised IT collaborators on proper star and snowflake schema database modeling and the impact of ONRR's database structure on its statistical projects and analyses.
 - Created numerous SQL queries for routine data export and the detection of aberrant production, royalty, and reference data anomalies.
 - Guided colleagues on the proper configuration of ONRR's various data connections (Oracle, Snowflake, etc.) and the initial setup and maintenance of each system's interfaces (SQL Developer, Snowflake Console, RODBC) and connectivity (ODBC, OCI).
 - Created data visualization products such as polished static explanatory images and interactive dashboards (Power BI, R Shiny package).
 - Used programmatic software (R, Alteryx, Power BI) to prepare data for analysis and cleaned imperfect data with statistical imputation and value matching.
- Leveraged understanding of regulations in *Federal Oil and Gas Royalty Management Act* and *Federal Oil and Gas Royalty Simplification and Fairness Act* for calculating and verifying correct royalty payment and creating corresponding models to automate these efforts.
- Mapped ONRR's preexisting outdated case tracking database structures and its numerous pitfalls, becoming a local expert in its organization and relation to front-end systems.

All positions listed in the EXPERIENCE section included at least 40 hours of work per week.

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- Led and coordinated the efforts of other ONRR colleagues and stakeholders in solving complex problem problems for a wide range of systems, applications, and customers.
- Served as *Executive Secretary* for Compliance Strategy Governance Board; performed agenda curation, meeting moderation, task tracking organization, and other administrative tasks to best empower the Board in effective strategic planning.
- Presented a training workshop *Machine Learning in R*, focusing on neural networks, the R statistical programming language, and predictive model training via gradient descent, at the 2024 DOI Economics Community of Practice Training Workshop.

University of Notre Dame, Notre Dame, IN

Analyst and Consultant, Office of Institutional Research

Winter 2020 - Spring 2022

- Provided analytical and statistical consulting services to numerous campus offices. Clients included HR, OIT, Office of Financial Aid, Office of the Provost, and college administrations.
- Led a team in reimagining the University's annual strategic plan assessment and in converting its 150 performance indicators from static presentations to interactive Tableau dashboards; this project included creating the backend pipelines and database infrastructure in Snowflake.
- Spearheaded the creation of a framework utilizing NSF and corporate data in benchmarking the representation of departmental faculty applicant pools to their corresponding hiring pools.
- Studied University student financial aid data, unearthed novel insights, created Tableau dashboards for leadership's self-guided exploration, and shared a statistical plan for optimizing aid spend to improve student attributes (graduation rate, extracurricular participation rate).
- Presented statistical underpinnings and methodology training for *Regression Diagnostics for Salary Impartiality Modeling* at 2022 INAIR conference for institutional research.

Associate Director of Data Analysis, Mendoza Graduate Programs

Fall 2017 - Winter 2020

- Managed over 80 analytics projects annually for wide-ranging groups of stakeholders including the College dean, associate deans, CMO, active recruiters, and admissions operations.
- Established statistical techniques for estimating prospect conversion probabilities, optimizing marketing click-through rates, and predicting graduate starting salary ranges.
- Developed extensive digital infrastructure in departmental CRM to improve data collection, streamline operations processes, and empower colleagues through self-service reporting.
- Instituted standardized A/B testing for email marketing campaigns and integrated this functionality into the CRM user interface for the Mendoza recruiting team.

1st Source Bank, South Bend, IN

Analyst, Cards and Electronic Payments

Fall 2014 - Fall 2017

- Orchestrated card usage campaigns, analyzed results, presented actionable statistical insights to senior management, and executed plan leading to a 90% increase in campaign success rate.
- Automated departmental reporting by researching all processes involved and producing applications for integration into dataflows, resulting in over 80 hours of weekly time savings.
- Statistically modeled estimated client churn (loss) for anticipated physical branch closures using customer segmentation coupled with regression tree approaches.
- Launched the Bank's inaugural data science initiative, leveraging the unique skills of team members from *Cards and Electronic Payments*, *IT*, *Accounting*, and *Operations*.
- Directed various departmental databases and educated teammates on use of software.

University of Saint Francis, Fort Wayne, IN

Visiting Assistant Professor, Department of Mathematics

Fall 2013 - Summer 2014

- Taught courses on *Statistical Applications*, *College Algebra*, and *Foundations of Mathematics*.
- Achieved very high teacher ratings, evidenced both by student success and by placing above the 90th percentile of student satisfaction, relative to all faculty across the University.

Valparaiso University, Valparaiso, IN

Graduate REU Coordinator, Department of Mathematics

Summer 2013

- Advised and mentored REU (Research Experience for Undergraduates) participants on mathematical research and in preparing papers, posters, and conference presentations in LaTeX.

University of Notre Dame, Notre Dame, IN

Research and Teaching Assistant, Department of Mathematics

Summer 2009 - Spring 2013

- Investigated research questions in algorithmic complexity with a multi-institutional group.
- Prepared and presented results at academic conferences (7 times) and seminars (6 times).
- Served as instructor for courses *Calculus I*, *Calculus III*, *Symbolic Logic*, *Axiomatic Set Theory* and created a RET (Research Experience for Teachers) program on *Theory of Infinite Games*.
- Achieved extremely high teacher ratings, evidenced both by student success and by placing above the 95th percentile of student satisfaction, relative to all faculty across the University.

RELEVANT SKILLS

Technology

- Highly proficient with programming using statistical software R, RStudio, Alteryx; expertise using the R Tidyverse and Tidy Modeling framework of packages.
- Experienced with writing complex SQL for queries, transformations, database object creation; Microsoft SQL Server, Oracle, and Snowflake environments, and their associated IDEs.
- Competent with efficient and effective data warehousing using star and snowflake schemas.
- Skilled with data visualization and dashboard building using Tableau and Power BI.
- Expert at LaTeX document typesetting and Beamer package for presentation building.

Statistics

- Theoretical foundations: probability theory, discrete distributions, the binomial distribution, Poisson distributions, chi-square distributions, normal distribution, central limit theorem.
- Applications: point estimation, confidence intervals for means/variances/proportions, statistical hypothesis testing (proportions, mean, variance, Wilcoxon, Chi-Square, etc.), outlier detection and model diagnostics (leverage, Cook distance, weight).
- Model types: (multiple) linear regression, (multiple) logistic regression, polynomial regression, regression splines, regression trees, classification trees, random forests, K -means clustering, hierarchical clustering, principal component analysis, feedforward neural networks, text mining (sentiment analysis, tf-idf, N-grams, document-term matrices, LDA topic modeling).
- Model workflows: pipeline creation, cross-validation, test set evaluation, feature engineering, resampling methods, model tuning, dimensionality reduction, blended model ensembles.
- Machine learning concepts: gradient descent, curse of dimensionality, bias-variance tradeoff, human interpretability, overfitting and cross-validation, cost function minimization.

Personality

- Excellent speaking, writing, and presentation skills, particularly for communicating technical content to broad audiences in industry, academia, and government environments.
- Extensive formal experience delivering complex course and presentation content in applied mathematical and computer sciences with superb effectiveness ratings.
- Energetic personality with a tenacious persistence for solving difficult, demanding problems.
- Proven record of dedication to developing and maintaining professional relationships.
- Well-versed in independent research and in mastering new concepts through self-study.